

NAME

kepler – two-player tactical space combat game command interface

SYNOPSIS

kepler [*command* [*arguments...*]]

DESCRIPTION

Kepler's Horizon is a two-player tactical game of space combat where players build individualized ships, move them across a hex-based star map, and resolve combat without dice. The objective is to occupy opponent base star hexes and accumulate victory points.

This manual describes all available commands in the Kepler's Horizon command language as defined by the game grammar.

GAME OVERVIEW**Map Components****StarHex**

A map hex containing a large or small star (graph node)

BaseStarHex

A map hex containing a large star (6 total: 3 per player)

SpaceHex

A map hex without a star or warpline

WarpLine

Bi-directional edges connecting stars for instantaneous travel

Ship Types**WarpShip (W#)**

Ships capable of interstellar travel via warplines

SystemShip (S##)

Ships that cannot travel between stars unless carried by WarpShips

GAME SEQUENCE

Each player-turn consists of the following phases in order:

1. Count Victory Points
2. Build Ships (receive BP in Advanced scenario)
3. Movement
4. Resolve Combat
5. SystemShip Pick and Drop
6. End of Turn

SCENARIOS**Learning Scenario**

40 BP initial, WarpShips only, 1 base star, first to 1 VP wins

Basic Scenario

50 BP initial, WarpShips and SystemShips, 1 base star, first to 2 VP wins

Advanced Scenario

20 BP initial + 10 BP/turn, all rules active, 3 base stars, first to 3 VP wins

COMMANDS

Session Management Commands**start learning**

Start a new game using the Learning scenario (40 BP, WarpShips only, 1 VP to win)

start basic

Start a new game using the Basic scenario (50 BP, WarpShips + SystemShips, 2 VP to win)

start advanced

Start a new game using the Advanced scenario (20 BP + 10/turn, all rules, 3 VP to win)

reset Reset the current game, wiping all game state

save name

Save the current game state under the specified name (alphanumeric identifier)

load List all saved games available for loading

load name

Load a previously saved game by name

delete Delete the most recently saved game

delete name

Delete a specific saved game by name

quit Quit the game, automatically saving as 'lastgame' (overwrites previous lastgame)

Information Commands

status Show game status and important records (current phase, turn number, BP, VP)

crt Display the Combat Results Table

hex location

Show location and spatial information about the specified hex

online Show which players are online and their connection times

stats Display statistics about the game, your fleet, and opponent

fleet Show manifest of your fleet (all ships and their current attributes)

system name

Show environmental information about the specified star system

galaxy Complete overview of all systems and all ships in the game

Turn Management Commands

next Advance the active player to the next phase

done Complete current player's turn and advance to opponent's first phase

Build Commands

build Show current build state (active drafts, available BP)

build new code name

Create a new ship draft with the specified code (e.g., W1, S01) and name

build drafts

List all pending build drafts

build drafts ship_code

Show detailed attributes of a specific draft ship

build set attributes

Set attributes on the current draft ship. Attributes are specified as:

pd=N or **d=N**

Power/Drive strength (1 BP per unit, provides movement and combat power)

b=N Beam weapon strength (1 BP per unit, max power for beam projector)

s=N Screen strength (1 BP per unit, absorbs hits equal to power + tech level)

t=N Number of missile tubes (1 BP per tube, requires 1 PD to fire)

m=N Number of missiles (1 BP per 3 missiles, each does 2 + tech level hits)

sr=N SystemShip racks (1 BP per rack, WarpShips only, carries 1 SystemShip per rack)

Example: build set pd=7 b=3 s=2 t=1 m=6

Note: Warp Generator costs 5 BP (automatically added for WarpShips)

build commit

Commit the current draft, spending BP and adding ship to your fleet

build cancel

Cancel the current draft without spending BP

Deployment Commands

deploy Show current deployment status for both players

deploy *ship_code system_name*

Deploy the specified ship to the specified base star hex (must be your base star)

Movement Commands

move Show current movement status for all ships

move *ship_code destination [waypoint...]*

Move the specified ship to destination, optionally through waypoints. Movement costs:

- 1 PD: Move to adjacent hex
- 1 PD: Enter warpline at star hex
- 1 PD: Travel full warpline to destination star
- 1 PD: Pick up one SystemShip
- 1 PD: Drop off one SystemShip

Ships must stop at star hexes occupied by enemy ships.

Ships may freely move through space hexes with enemy ships.

Combat Commands

combat

Show status of current combat situation

combat order *ship tactic target allocation*

Issue combat orders for a ship. Components:

ship Your ship identifier (e.g., W1, S05)

tactic Combat tactic: **attack**, **dodge**, or **escape** (WarpShips only)

target Enemy ship identifier to target

allocation

Power allocation from PD to attributes:

pd=N or **d=N**

Drive allocation (for maneuver, affects CRT row/column)

b=N Beam power (cannot exceed built capacity)

s=N Screen power (cannot exceed built capacity)

t=N Tubes to power (1 PD per tube)

m=N Missiles to fire (requires powered tubes)

Note: Total allocation cannot exceed current PD

Note: Cannot use Beams/Screens and Missiles in same round

combat order commit

Commit all combat orders for this round

combat order cancel

Cancel all combat orders and re-issue

combat apply ship damage_allocation

Apply damage hits to ship attributes (defender chooses where hits are taken)

combat apply commit

Commit damage allocation

combat apply cancel

Cancel damage allocation and re-apply

Pick and Drop Commands

pick location

Show status of what can be picked up at the specified hex

pick systemship warpship

Pick up the specified SystemShip into the specified WarpShip's rack

drop warpship

List what may be dropped from the specified WarpShip

drop systemship warpship

Drop the specified SystemShip from the specified WarpShip at current location

Repair and Resupply Commands

repair List ships that can be repaired (must be at your base star hex)

repair ship_code

Repair the specified ship using BP (1 BP repairs 1 unit of any attribute up to original max)

resupply

Show available BP and resupply capabilities (missiles only)

resupply ship_code quantity

Resupply the specified ship with missiles (1 BP per 3 missiles, up to original max)

Help Commands

help Show help screen and list all help topics

help topic

Show help about a specific topic or command

SHIP ATTRIBUTES

Power/Drive (PD)

Engine strength. Provides movement points and combat power allocation. Cost: 1 BP per unit.

Warp Generator (WG)

Enables interstellar travel and warpline usage. Cost: 5 BP. Required for WarpShips.

Beams (B)

Energy weapon projector. Max strength = BP spent. Cost: 1 BP per unit. Damage = power + tech level.

Screens (S)

Protective energy screen. Absorbs hits = power + tech level. Cost: 1 BP per unit.

Tubes (T)

Missile launchers. Each tube fires 1 missile/round. Requires 1 PD to power. Cost: 1 BP per tube.

Missiles (M)

Projectile weapons. Damage = 2 + tech level. Cost: 1 BP per 3 missiles.

SystemShip Racks (SR)

Carries SystemShips (WarpShips only). 1 SystemShip per rack. Cost: 1 BP per rack.

TECHNOLOGY LEVELS

In Advanced scenario, tech level increases every 4 turns:

Turns 1-4: Level 0
 Turns 5-8: Level 1
 Turns 9-12: Level 2
 Turns 13-16: Level 3
 (continues +1 per 4 turns)

Tech level adds to:

- Beam damage when hitting
- Missile damage when hitting (2 + tech level)
- Screen absorption when powered

COMBAT RESOLUTION**Combat Results Table (CRT)**

The CRT determines hit/miss based on:

- Firing ship's tactic (Attack/Dodge/Retreat)
- Target ship's tactic (Attack/Dodge/Retreat)
- Drive difference (Firing Drive - Target Drive)

Combat Tactics

Attack Aggressive offense. Best hit chance when drive advantage is 0 to +2.

Dodge Defensive maneuvering. Good hit chance at +1/+2 drive, hard to hit at 0/-1.

Retreat (WarpShips only)

Attempt to escape combat. Must get "Escapes" result against all enemy ships.

Damage Application

1. Calculate hits from all weapons that hit
2. Subtract Screen power + tech level (if Screen powered)
3. Apply remaining "effective hits" to ship attributes (owner chooses where)
4. Ship destroyed when all attributes except Warp Generator reduced to 0

Special Combat Rules

- Beams and Screens can be used together
- Missiles cannot be used same round as Beams/Screens
- Missiles always use Attack tactic with their own drive setting
- SystemShips cannot Retreat (must be carried by WarpShip)
- Combat ends when: all ships of one side destroyed/retreated, or 3 consecutive rounds with no unabsorbed damage (attacker must withdraw)

MOVEMENT RULES

1. WarpShips must stop at star hexes occupied by enemy ships
2. WarpShips may move through space hexes with enemy ships
3. WarpLines are treated as space hexes for movement
4. Ships with PD=0 cannot move
5. PD used for movement is not expended (full PD available for combat)
6. Cannot move onto enemy base star hex during turn 1

VICTORY CONDITIONS**Dominance Victory**

Accumulate required victory points (1/2/3 depending on scenario) by occupying enemy base stars at start of your turn

Elimination Victory

Destroy all enemy ships

Draw Neither player has effective ships remaining (Learning/Basic scenarios only)

BUILD POINT COSTS

Attribute	Cost
Power/Drive (PD)	1 BP per unit
Warp Generator (WG)	5 BP
Beam (B)	1 BP per unit
Screen (S)	1 BP per unit
Tube (T)	1 BP per tube
Missiles (M)	1 BP per 3 missiles
SystemShip Rack (SR)	1 BP per rack

EXAMPLES**Building a WarpShip**

- > build new W2 Destroyer
- > build set pd=5 b=3 s=2 t=1 m=3
- > build commit

Total cost: 5+3+2+1+1+5(WG) = 17 BP

Tech level 0 (if built turns 1-4)

Building a SystemShip Fighter

- > build new S55 Fighter
- > build set pd=1 t=1 m=6
- > build commit

Total cost: 1+1+2 = 4 BP

No Warp Generator (SystemShip)

Movement Example

> move W6 Girsu Kish

W6 moves from current hex to Girsu (1 PD), along warpline to next star (1 PD), then 3 hexes to Kish (3 PD) = 5 PD total

Combat Order Example

> combat order W3 attack S25 d=0 b=3 s=2

W3 attacks S25, allocating 0 to drive, 3 to beam, 2 to screen (requires PD¥5)

Missile Fire Example

> combat order S25 dodge W3 d=4 t=1 m=1

> (missile drive setting: 3)

S25 dodges while firing 1 missile at W3 with drive=3 (requires PD¥5)

CONSTRAINTS AND INVARIANTS

- No randomness - all outcomes deterministic
- No hidden information - all game state is public
- Rules do not change mid-game
- Equal inputs yield equal outcomes
- Ships retain original tech level permanently
- Warp Generators never take damage (ship destroyed when all else is 0)
- SystemShips cannot have Warp Generators or SystemShip Racks
- WarpShips cannot be carried in SystemShip Racks

REPAIR AND RESUPPLY

- Ship must start turn at your base star hex
- SystemShips can be repaired/resupplied while loaded on WarpShip
- 1 BP repairs 1 unit of any attribute
- Cannot repair beyond original built capacity
- Tech level never changes
- 1 BP resupplies up to 3 missiles
- Cannot resupply beyond original missile capacity

SYSTEMSHIP PICKUP/DROP

During Movement

- Costs 1 PD per SystemShip picked up or dropped
- Can be done at any star hex during movement

During Combat

- Requires: Drive=0, Screen=0, Dodge or Retreat tactic

- Only 1 SystemShip per combat round
- May fire Beam (but not Missiles) while picking up/dropping
- Dropped SystemShips cannot fire that round
- Picked up SystemShips cannot fire but may power Screens

After Combat

- Free rearrangement of SystemShips at star hexes
- No movement cost
- Active player only

FILES

~/kepler/saves/

Directory containing saved game files

~/kepler/lastgame

Automatically saved game when quitting

SEE ALSO

RULES.md(5), **COMBAT_RULES.md(5)**, **DESIGN.md(5)**

NOTES

- All ship records must be maintained by players
- Records are shown to opponent after (not during) game
- In Advanced scenario, track BP received and expended
- Combat orders are written simultaneously and revealed together
- Fractions of BP from missile resupply are not saved

BUGS

Report bugs to the Kepler's Horizon issue tracker.

AUTHORS

Game design and implementation by sibomots.

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GAME DESIGN PHILOSOPHY

Kepler's Horizon is intentionally minimal. Strategic depth emerges from:

- Constrained movement
- Simultaneous decision-making
- Irreversible consequences

The rules favor clarity over complexity.